

Subspace Alignment for Vision-Language Model Test-time Adaptation

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BACKGROUND

➤ VLM test-time adaptation

- ❖ Adapt pretrained VLM to shifted unlabeled test data without re-accessing source data or labels.

➤ Existing works: reliance on zero-shot predictions

- ❖ **Training-free**: cache confident zero-shot predictions.
- ❖ **Training-based**: optimize TTA loss based on predictions.

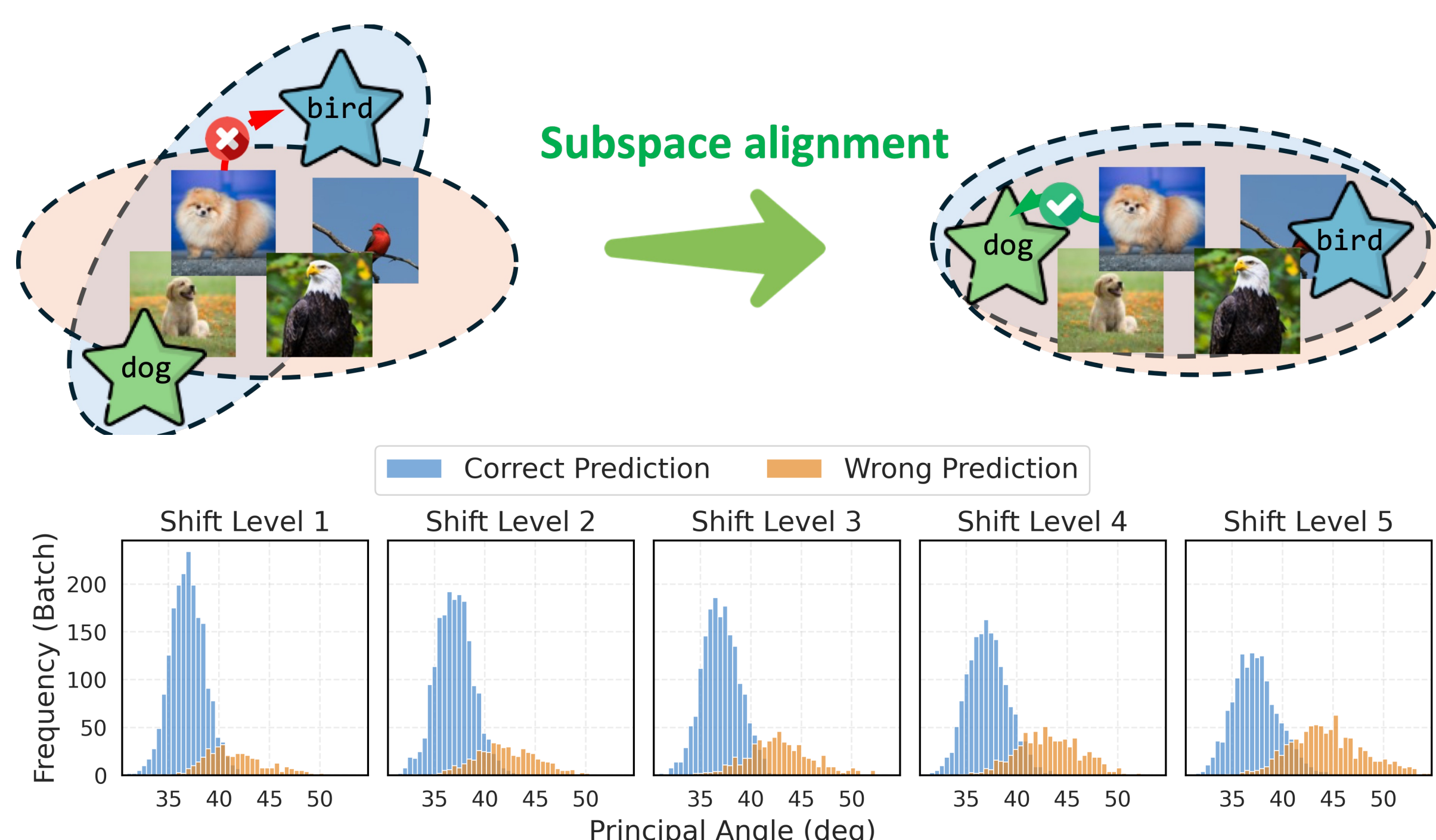
➤ Can we trust zero-shot predictions?

- ❖ Inaccurate predictions can reinforce errors.

OBSERVATIONS

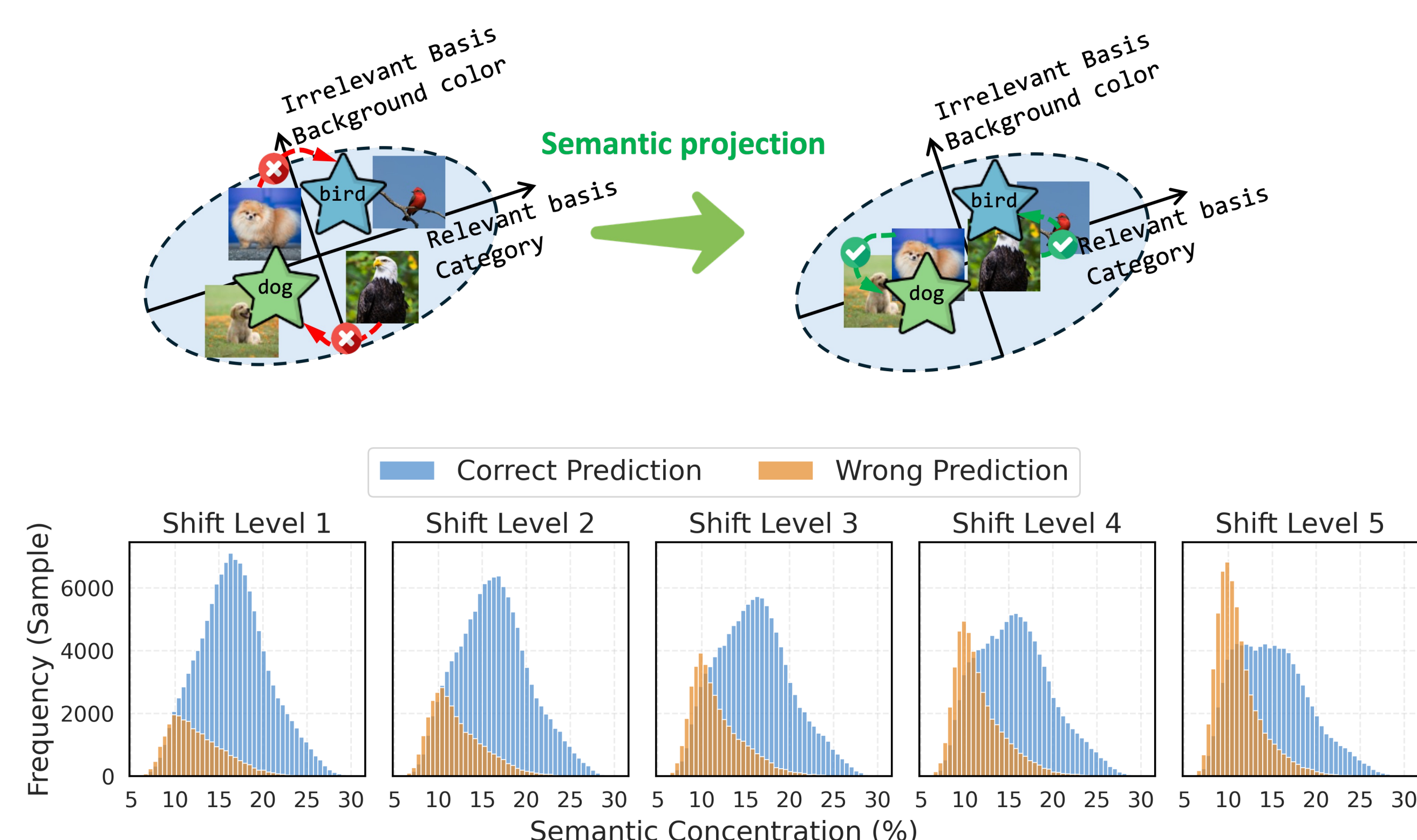
➤ 1. Modality gap → Geometric alignment

- ❖ **Cause**: misaligned visual-textual space shifts visual (dog) features toward incorrect text (bird) anchor.
- ❖ **Observation**: Shifts exacerbate the gap (principal angle) between two modalities.



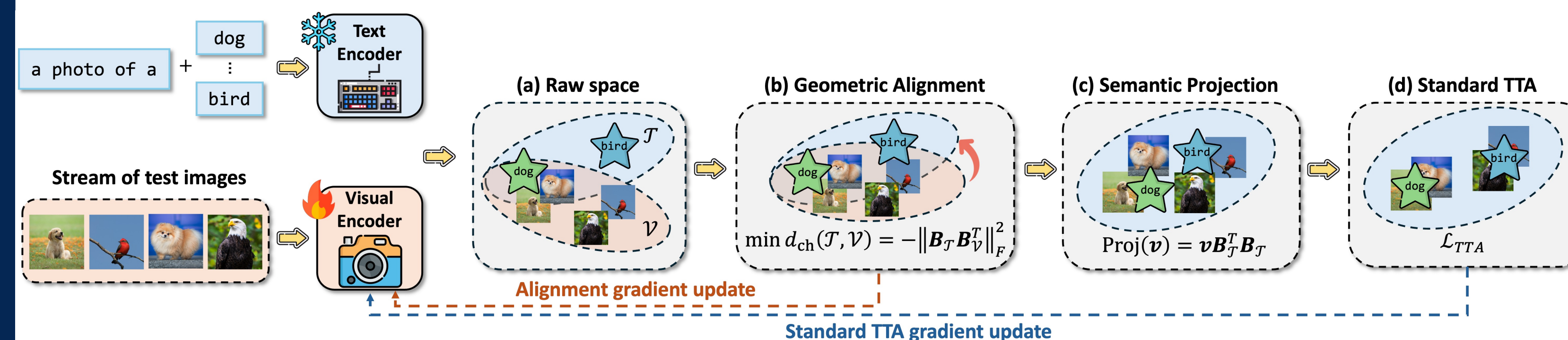
➤ 2. Visual nuisance → Semantic projection

- ❖ **Cause**: misaligned visual-textual space amplifies the irrelevant bases (color) while suppress relevant information (animal).
- ❖ **Metric**: Shifts degrade semantic concentration of visual embeddings



METHODOLOGY

➤ SubTTA: Align → Project → Adapt



➤ 1. Geometric alignment

- ❖ **Subspace extraction**: top-r bases of normalized features.
- ❖ **Moving average**: stabilize small batches and class coverage.

$$\Sigma_T = T^T T = B_T^T \Lambda_T B_T$$
$$\Sigma_V \leftarrow (1 - \alpha) \Sigma_V + \alpha V^T V = B_V^T \Lambda_V B_V$$

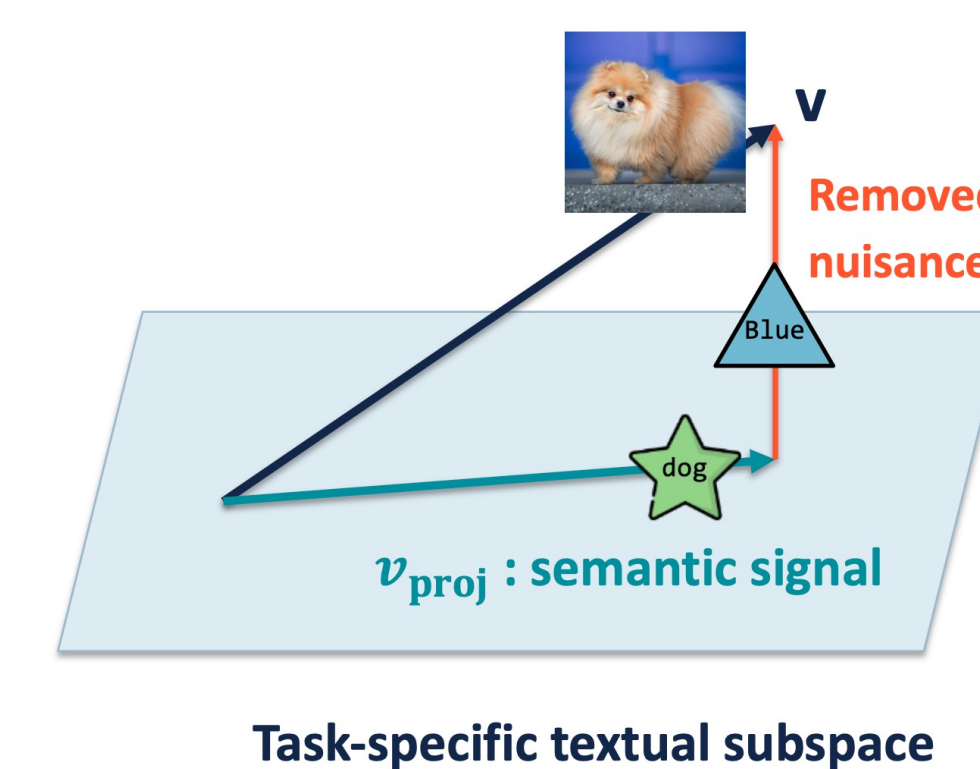
- ❖ **Geometric alignment**: update visual encoders to align with frozen textual bases

$$\mathcal{L}_{\text{align}} = r - \|B_T B_V^T\|_F^2$$

➤ 2. Semantic projection

- ❖ **Retain**: task semantics within the textual span.
- ❖ **Filter**: nuisance orthogonal to the span.

$$v_{\text{proj}} = v B_T^T B_T$$



➤ 3. Standard test-time adaptation

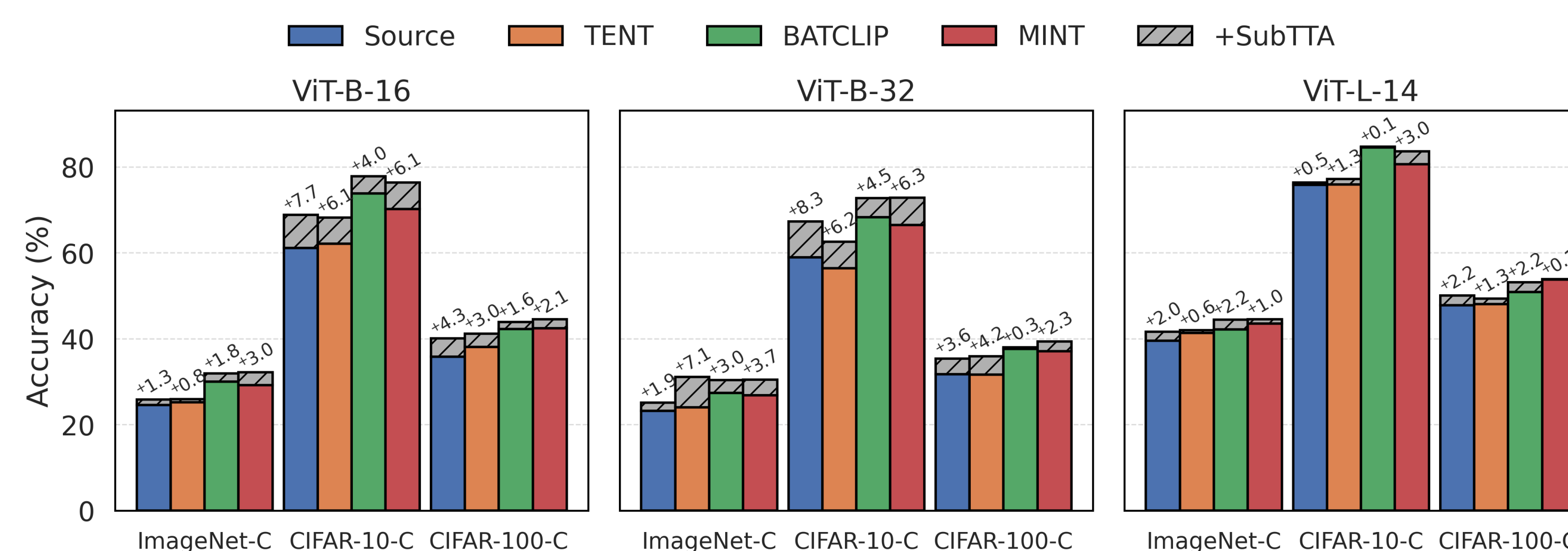
- ❖ Adapt on projected features with entropy / cluster objectives.
- ❖ Compatible with TENT, BATCLIP and MINT.

EXPERIMENTS

➤ Experiment Setup

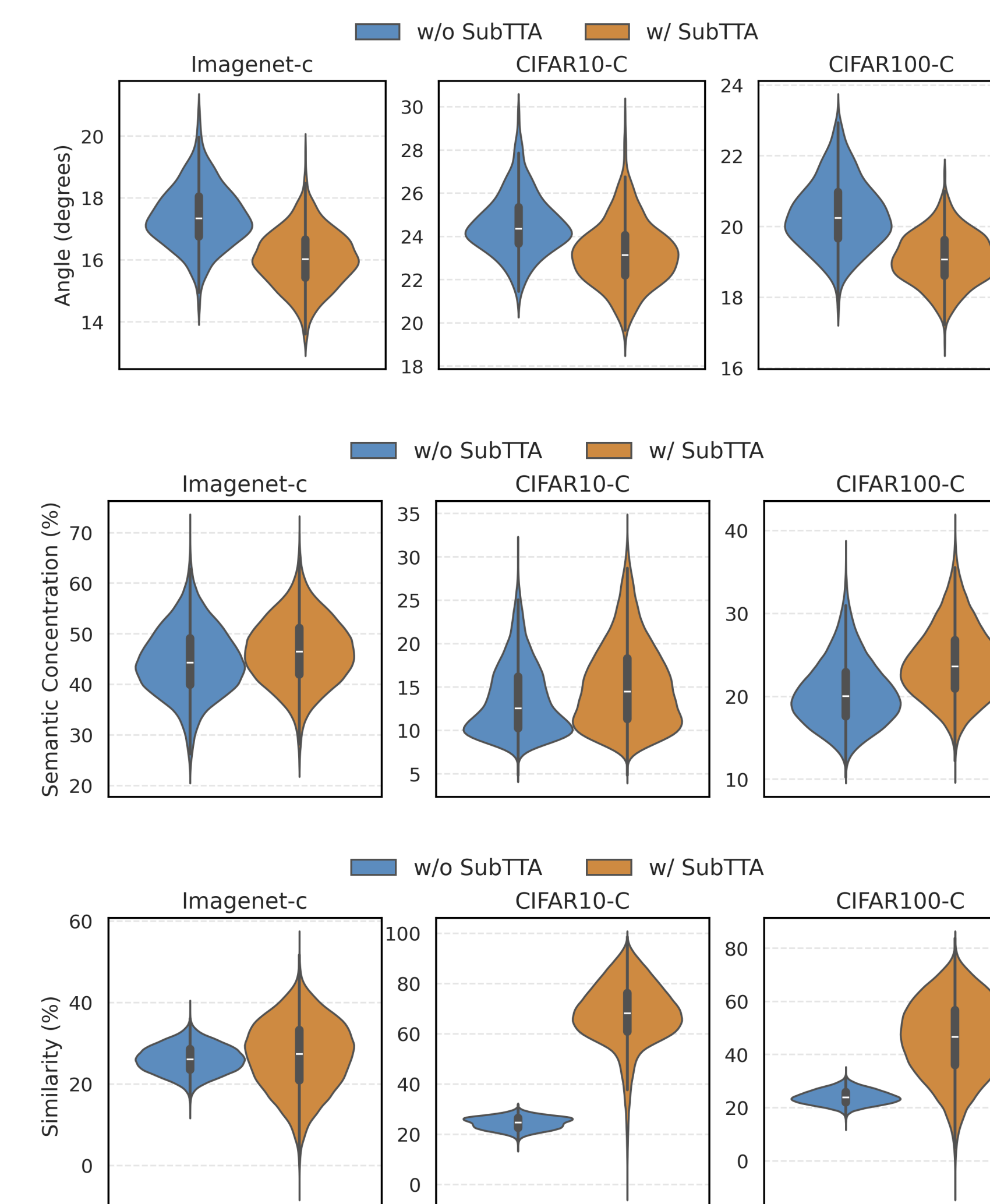
- ❖ Datasets: ImageNet-C / CIFAR-10-C / CIFAR-100-C.
- ❖ Models: CLIP ViT-B-16 / B-32 / L-14.

➤ Benchmark Results



- ❖ Consistent improvement over diverse baselines.
- ❖ Gain up to +8.3 pp; Average gain +2.24 pp.
- ❖ Avoids negative adaptation and ensures stability

➤ Embedding quality



➤ Principal angles ↓

- ❖ Reduce the modality gap.

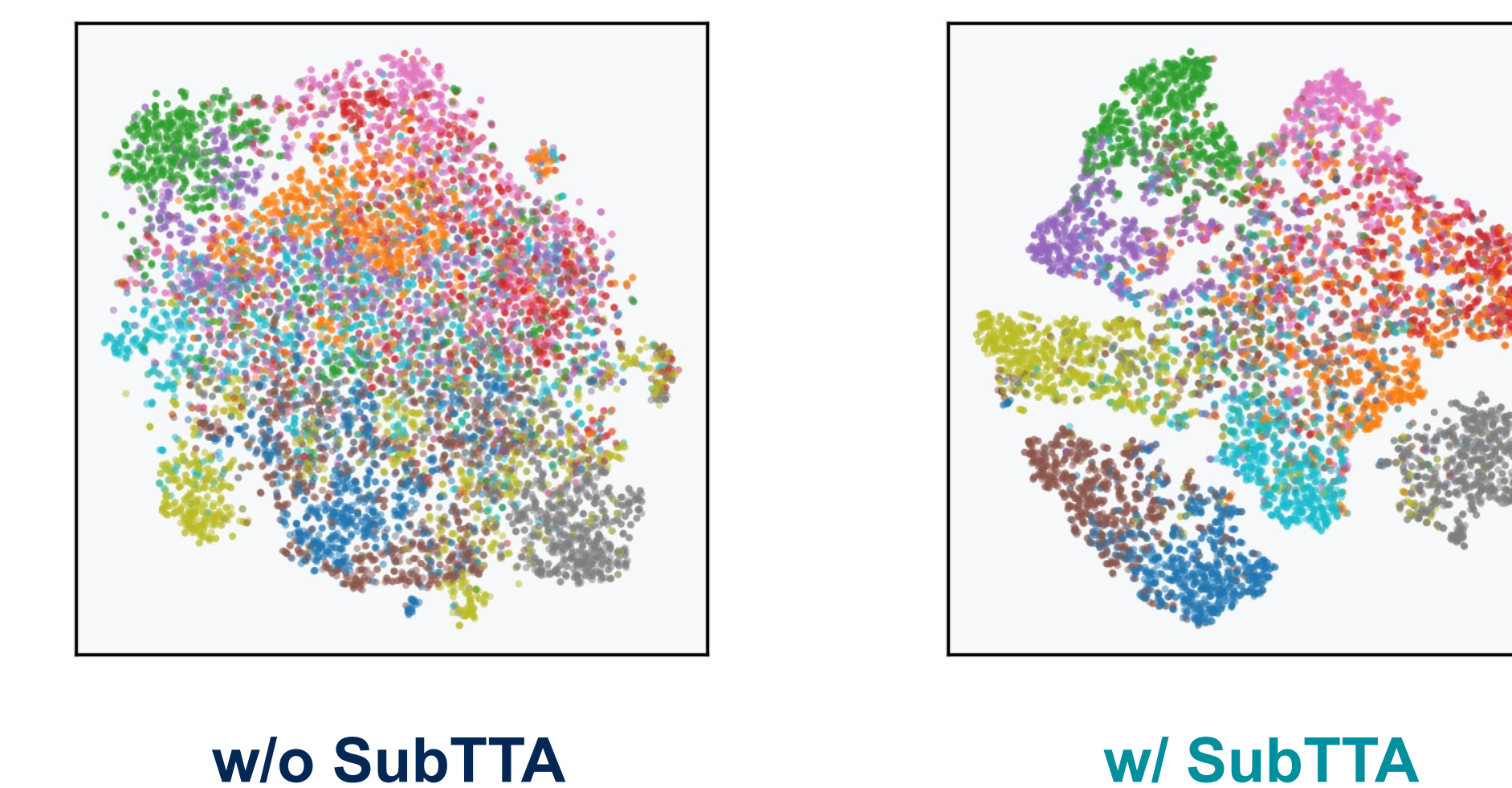
➤ Semantic concentration ↑

- ❖ Preserve task-specific semantics.

➤ True-label similarity ↑

- ❖ Improve visual-text matching.

➤ Embedding visualization



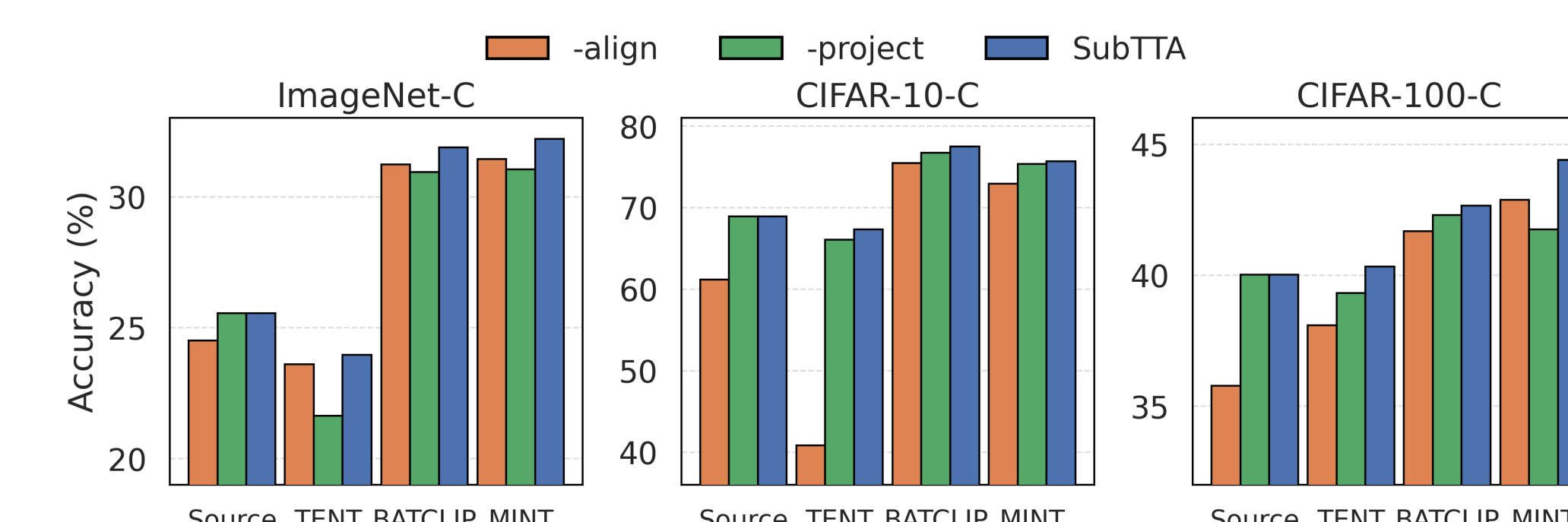
➤ Improved compactness

- ❖ Tighter within-class clusters.

➤ Improved separation

- ❖ Clearer class boundaries.

➤ Component study (ablation)



- ❖ **Without alignment**: projection may amplify errors.

- ❖ **Without projection**: visual nuisance remains.

ACKNOWLEDGEMENTS



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Paper



Code



Slides



Poster

